

Poly(A)-site clustering reproduces gene-expression cell types on public PBMC data.

The signal is 3'-end expression, not poly(A)-site choice, and no PAS-only population survives.

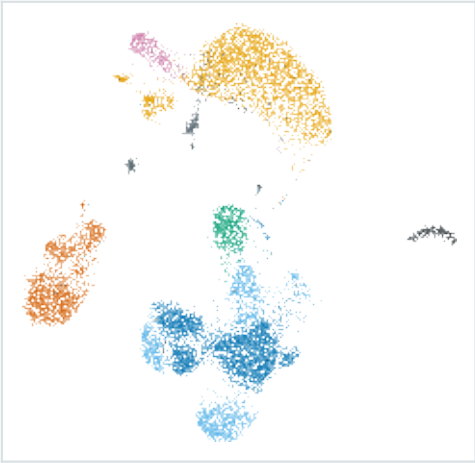
10x pbmc_10k_v3 (11,836 PAS cells / 11,043 GEX cells; 11,041 matched, 93.3% of PAS cells). PeakATail PAS calls, TF-IDF/LSI/Leiden on PAS counts only. Independent replication of the Laughney-cohort result.

a GEX-derived embedding



10x gene counts → HVG/PCA/Leiden; colour = marker-derived label

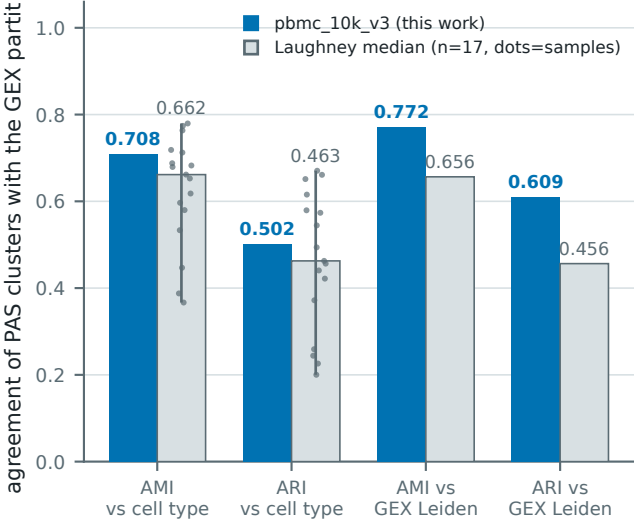
PAS-derived embedding



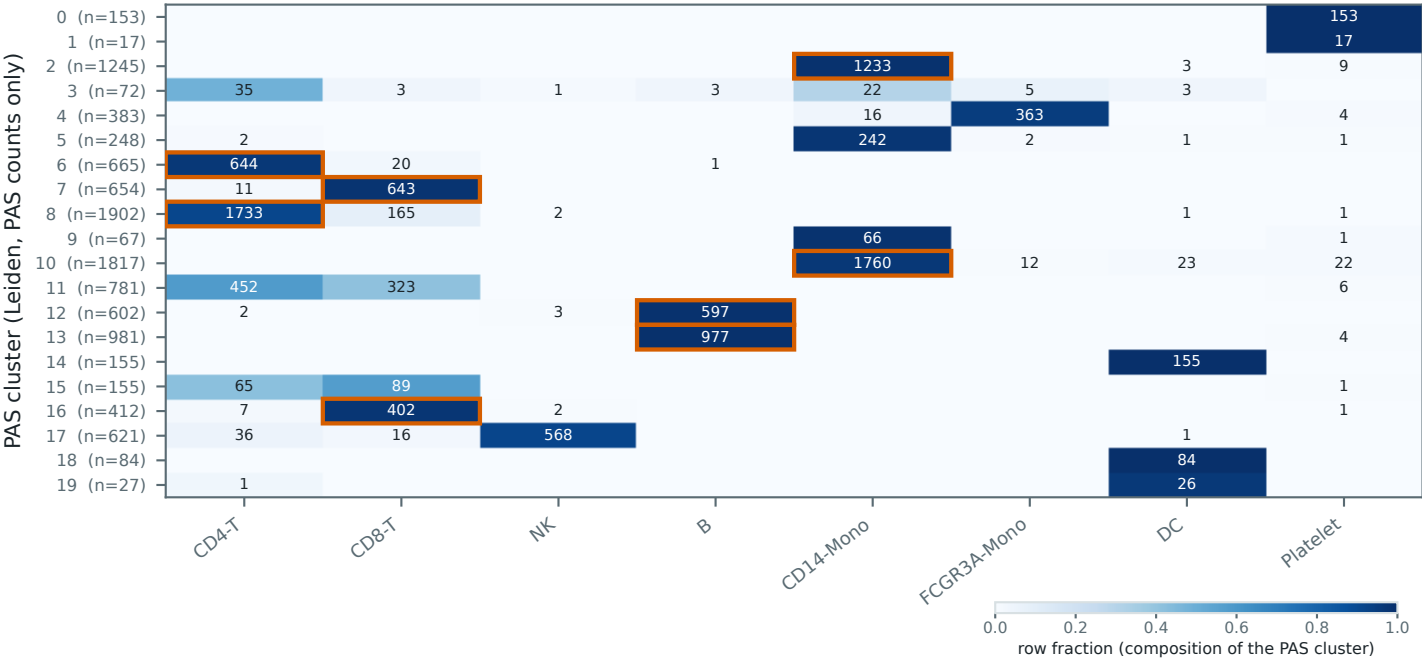
275,370 PAS counts → TF-IDF/LSI/Leiden; no gene expression used

CD4-T (2988) NK (576) CD14-Mono (3339) DC (297)
CD8-T (1661) B (1578) FCGR3A-Mono (382) Platelet (220)

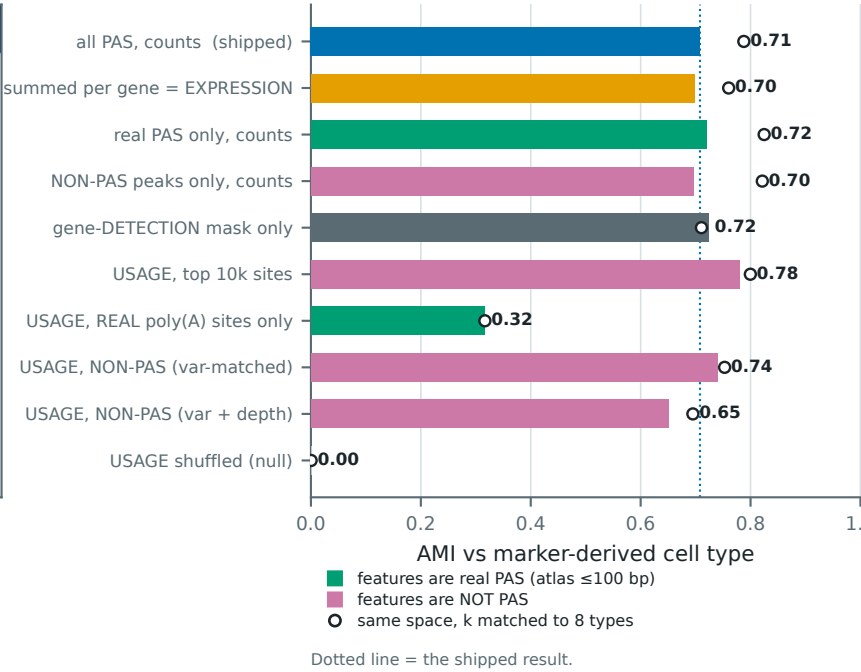
c the Laughney claim replicates



b PAS cluster × marker-derived GEX type (orange = the four detected splits)



d what the PAS space actually encodes identical pipeline, one kind of information removed each time; Leiden resolution 1.0



e novelty test — does any PAS-only split survive? Pre-registered criteria, applied in order

	depth (PAS UMI)	depth (GEX UMI)	mito %	doublet score	ambient / off-lineage	plain expression	GEX Leiden already splits	discriminating sites are real PAS
B PAS 13 (n=977) vs PAS 12 (n=597)	1.4× AUC 0.20	1.5×	1.0×	1.4×	1.4× Mono	18 DE genes	ARI 0.90	12/73 =16%
CD14-Mono PAS 10 (n=1760) vs PAS 2 (n=1233)	2.2× AUC 0.97	2.0×	1.1×	1.3×	1.0× Platelet	0 DE genes	ARI 0.09	4/10 =40%
CD4-T PAS 8 (n=1733) vs PAS 6 (n=644)	1.1× AUC 0.65	1.1×	1.0×	1.5×	1.0× Mono	16 DE genes	ARI 0.59	5/51 =10%
CD8-T PAS 7 (n=643) vs PAS 16 (n=402)	1.0× AUC 0.52	1.1×	1.1×	1.9×	2.6× NK	24 DE genes	ARI 0.80	0/33 =0%

VERDICT: no split survives. Every one of the four PAS-only splits fails at least one pre-registered criterion. Pooled across splits, 167 PAS pass the APA-driven filter, but only 21 (12.6%) are within 100 bp of a PolyASite 2.0 site — significantly FEWER than the 20.0% of the sites they were drawn from (Fisher OR 0.54, p = 0.009), and no better than the 11.8% background of all PeakATail PBMC calls. The 'APA-driven' signal is carried by peaks that are not poly(A) sites.

split: ≥2 PAS clusters each ≥60% one type, each ≥10% of it and ≥100 cells | candidate site: Δusage| ≥ 0.2, gene ≥100 counts/arm | significant: per-CELL usage, Mann-Whitney, BH q<0.05
APA-driven: significant AND |log2FC| ≤ 0.5 for the gene's TOTAL expression in the independent 10x GEX matrix | confound flagged at >1.5× median ratio, expression at ≥10 DE genes (q<0.05, |log2FC|>1), GEX-resolved at ARI ≥ 0.2

CAVEATS ON EVERY NUMBER ABOVE. (1) The reference partition is MARKER-DERIVED, not curated ground truth: canonical PBMC marker panels scored per GEX-Leiden cluster, modal label; every error of that panel is inherited here. (2) The shipped PAS features are per-site COUNTS, which carry gene abundance; panel d shows that collapsing them to gene totals costs nothing, so the concordance is a 3'-end EXPRESSION result and is not evidence of isoform choice. (3) On this exact dataset PeakATail is LAST among de novo callers for PAS accuracy (F1 0.134 vs polyApipe 0.261) and only 11.8% of its 275,370 PBMC sites lie within 100 bp of a PolyASite 2.0 representative site; that is why the atlas column of panel e and the real-PAS bars of panel d are decisive. (4) Absence of surviving splits is not proof that none exist: splits are scored on the two largest type-dominated PAS clusters of each type, and the atlas is itself incomplete. (5) One dataset, one donor, one parameter set. (6) In panel d the two continuous usage spaces are each shown at the better of their raw and column-scaled representation, so the collapse of the real-poly(A)-site bar is not an artefact of representation; the open circles repeat every space with its Leiden resolution returned so the cluster count matches the 8 cell types, which removes granularity as an explanation. (7) Real poly(A) sites are sparser here (3 sites/gene, gene detected in 3.4% of cells) than PeakATail's non-PAS peaks (10 sites/gene, 13.2%), so the last bar repeats the non-PAS decoy after binomially thinning its counts to the same per-gene depth (4.0% detection, median gene total 957 vs 956): it still doubles the real-PAS score, so measurement power is not the explanation.